H-B-1654
June 7, 1971
SUPERSEDING
MIL-B-43363
September 9, 1965

FEDERAL SPECIFICATION

BRUSHES, DUSTING, LENS AND PHOTOGRAPHIC NEGATIVES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of lens dusting brush (see 6.1).

1.2 Classification. The lens dusting brushes covered by this specification shall be of the following sizes as specified (see 6.2):

Size - 1 inch width.
Size - 3 inch width.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

R-N-91	-	Naphthalene, Technical.
UU-P-268	-	Paper, Kraft, Untreated, Wrapping.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.
PPP-B-665	-	Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676	-	Boxes, Set-Up, Paperboard.

Federal Standard:

Fed. Std. No. 191 - Marking for Domestic Shipment (Civil Agencies).

FSC 7920

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, D.C. Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification

(Application for copies shall be addressed to the National Motor Freight Traffic Association, Inc., Agent, 1616 P Street, N.W., Washington, D.C. 20036.)

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Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

American Society for Testing and Materials (ASTM)

D 2016 - Standard Methods of Test for Moisture Content of Wood

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The brushes delivered under this specification shall be the manufacturer's standard commercial product, except as otherwise specified herein.

3.2 Materials. Materials shall conform to the requirements specified hereinafter. Materials not definitely specified shall be of the quality used by the manufacturer provided the item complies with all the provisions of this specification.

3.2.1 Hair. At the option of the supplier hair shall be camel hair (natural squirrel tail hair), or the hair shall be a blend of soft animal hairs consisting of 40 percent pony hair and 60 percent ox ear hair (see 4.2.1.1). All hair shall be straight resilient and new. The camel hair or the blend of pony hair and ox ear hair, as applicable shall be blended to a uniform color, cleaned, and sterilized as specified in 3.4. The hair shall show no evidence of artificial coloring when tested as specified in 4.2.2.3.

3.2.2 Wood. Wood shall be close-grained and straight-grained hardwood thoroughly and uniformly dried to a moisture content of not more than 9 percent and not less than 5 percent when tested as specified in 4.2.2.2.

3.3 Construction. The brushes shall be similar to figure 1 (see 6.3) and shall conform to the requirements specified herein and in table I. Ferrules shall be of metal with a minimum thickness of 0.015 inch and shall be lockseamed, soldered, or welded in accordance with the manufacturer's standard practice. Handles shall be fabricated from the wood specified in 3.2.2. The hair specified in 3.2.1 shall be set in vulcanized hard rubber or setting compound

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and shall break before separating from the rubber or setting compound when tested as specified in 4.3.1. The handles, ferrules and vulcanized hard rubber or setting compound containing the hair shall be fastened together with escutcheon pins, rivets or staples in accordance with the manufacturer's standard practice.

TABLE I. Physical requirements

Characteristics		One inch width	Three inch width
Overall dimensions			
Length	inches	6 ± 1/2	8 ± 1/2
Width	inches	1 ± 1/16	3 ± 3/32
Thickness	inch	1/4 ± 1/64	1/2 ± 1/16
Handles			
Length out of ferrule	inches	4 ± 1/2	5 ± 1/2
Width of shoulder	inches	1 ± 1/32	3 ± 1/32
Thickness	inch	1/4 ± 1/32	1/2 ± 1/16
Ferrules			
Depth	inches	1-1/4 ± 1/4	1-1/4 ± 1/4
Hair			
Weight, minimum (see 4.2.1.1)	ounce	1/8	1/2
Trim length out of ferrule, minimum	inches	1	1-1/2

3.4 Sterilization. The hair shall be sterilized, after thorough washing and rinsing, in accordance with any of the procedures specified in 3.4.1, 3.4.2 or 3.4.3. A recording device to provide a record of the pressures and temperatures, shall be provided.

3.4.1 Boiling water. The hair shall be soaked in boiling water (212° F.) for not less than 30 minutes.

3.4.2 Dry heat. The hair shall be placed in a chamber and a temperature above 284°F. shall be maintained for not less than 3 hours.

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3.4.3 Autoclaving. The hair shall be placed in a steam pressure chamber and subjected to a minimum temperature of 248°F., at 15 p.s.i.g, for not less than 45 minutes.

3.5 Finish. Wood handles shall be coated with varnish, lacquer, or other clear organic material in accordance with the manufacturer's standard commercial practice.

3.6 Marking for identification. The supplier shall permanently and legibly indent his name, or trademark of such known character, easily identifiable with said manufacturer, on each brush, in addition to any special markings required by the contractor or order (see 6.2).

3.7 Workmanship. All wood shall be smoothly finished and free of torn or loose grain, all corners shall be broken and all edges shall be rounded. All metal shall be free of slivers, burrs, and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.1.1 Certification of compliance. Unless otherwise specified in the contract or order, a certificate of compliance from the contractor may be accepted for the weight requirement specified in table I, the kind of hair specified in 3.2.1, and, when applicable, the percentage of each kind of hair specified in 3.2.1.

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4.2.2 In-process inspection.

4.2.2.1 In-process examination. Examination shall be made during the manufacturing process to determine that the sterilization of the hair is in accordance with 3.4, and, when applicable, the blend of 40 percent pony hair and 60 percent ox ear hair is in accordance with 3.2.1 In addition, natural squirrel tail hair or the blend of pony hair and ox ear hair, as applicable, shall be examined for color uniformity. Whenever nonconformance is noted, correction shall be made to the process and items affected.

4.2.2.2 Moisture content of handles. The handles shall be tested for moisture content prior to assembly and finish in accordance with any method of ASTM D2016, to determine compliance with 3.2.2. The sample unit shall be one handle. The lot shall be the number of handles of one size offered for inspection at one time. Three determinations shall be taken on each sample unit and the results shall be the average of the three determinations. The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.5 expressed in terms of defects per hundred units.

4.2.2.3 Color of hair. Before sterilization and prior to setting in the compound the hair shall be tested for evidence of artificial coloring. The sample unit shall be one ounce of hair. The lot shall be all the hair offered for inspection at one time. The sample shall be boiled in 8 ounces of clear water for 30 minutes adding replacement water as required. The water shall then be examined to determine compliance with 3.2.1. Any evidence of artificial coloring shall be cause for rejection of the lot.

4.2.3 End item inspection. The lot shall consist of all brushes of one size offered for inspection at one time. The sample unit shall be one completely fabricated brush.

4.2.3.1 Visual examination. Examination shall be made of the end item for the defects listed in table II. The inspection level shall be level II with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE II. Visual defects

Examine	Defect
Finish	Wood components not finished with transparent varnish, lacquer, or other clear organic material.
Construction and workmanship	Not design specified. Not materials specified.

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TABLE II. Visual defects (con.)

Examine	Defect
Construction and workmanship (cont'd)	Filler material not fastened with escutcheon pins, rivets, or staples. Filler material not clean. Filler material not trimmed to specified length. Handle not smoothly finished. Splintered, torn, or loose grain. Ferrules not lockseamed, soldered, or welded. Sharp edges, burrs, slivers, or metal splinters on ferrules. Component missing.
Marking for identification	Omitted, illegible.

4.2.3.2 Dimensional examination. Inspection shall be made of the end item for compliance with table I. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.2.3.3 Testing of the end item. The end item shall be tested as specified in 4.3.1 and 4.3.2. Any failure shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.2.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and marking comply with the requirements of section 5 of this specification. Defects will be scored as listed in table III. The lot shall be the number of containers offered for delivery at one time. The sample unit shall be one shipping container fully prepared for delivery except it need not be sealed. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units. Examination for closure and strapping shall be in accordance with applicable box specification.

TABLE III. Examination of preparation for delivery

Examine	Defect
Markings (exterior and interior)	Missing; incorrect; illegible; of improper size, location, sequence or method of application.

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TABLE III. Examination of preparation for delivery (con.)

Examine	Defect
Materials	Component missing, damaged, or otherwise defective.
Contents (exterior and interior)	Content per container is more or less than required.

4.3 Tests.

4.3.1 Set retention. Single strands of hair shall be pulled in a direction tending to pull them from the rubber or setting compound. The strands shall be pulled until they break or separate from the rubber or compound to determine compliance with 3.3. Ten determinations shall be made per unit. Any nonconformance with specified requirements shall constitute failure of this test.

4.3.2 Hair weight. Prior to setting the hair in the setting compound of the applicable size brush, the hair, the brush handle, the ferrule, and setting compound shall be weighed separately and recorded. The brush shall then be completely assembled and the hair trimmed to length. The finished brush shall be weighed. A determination shall then be made for compliance with 3.3 and table I. Any nonconformance with the specified requirements shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each brush shall be individually wrapped in a double thickness of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. The ends of the kraft paper shall be folded back over each end of the brush and secured with tape. As an alternate, the hair end (including the ferrule) of the brush shall be encased in a wallet-type envelope of 60-pounds minimum basis weight kraft paper or equivalent strength paper and secured to the handle of the brush with attached tie strings or tape.

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5.1.1.2 Intermediate package. Twelve brushes of one size only, unit packaged as specified in 5.1.1.1 shall be packaged in a snug-fitting folding, set-up or metal-stayed box conforming to variety 1, style III or VIII, type G, class "1" of PPP-B-566 or type I, variety 1, class A, style 3 or 4 of PPP-B-676; or class 1, style A of PPP-B-665, respectively. The brushes shall be arranged in an alternating hair to handle order. Not less than 3 grams of naphthalene in ball or flake form conforming to R-N-91 shall be placed in each box. Closure shall be in accordance with the appendix of the applicable box specification.

5.1.2 Level C. Brushes shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice, when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- One hundred and forty-four, size - 1 inch width brushes, or seventy-two size 3-inch width brushes, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to grade V2s style RSC of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with strapping or tape banding shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B. One hundred and forty-four, size 1-inch width brushes, or seventy-two size 3-inch width brushes, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 200 of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.2.3 Level C. Brushes, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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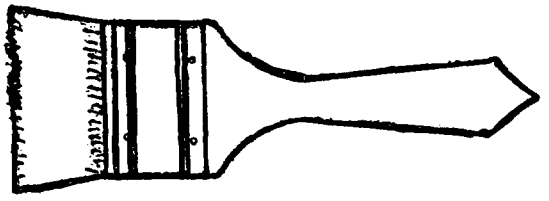


Figure 1 -- Lens Dusting Brush

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5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED. STD. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The brushes covered by this specification are intended for use by photographers for dusting photographic lenses and negatives.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) Size of brush required (see 1.2).
(c) Special marking when required (see 3.6).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Illustration. The illustration shown herein as figure 1 is for the purpose of identification and is not intended to preclude any brush which otherwise meets the requirements of this specification.

6.4 Method of packaging referred to in 5.1.1.2, is similar to method III of MIL-P-116.

MILITARY CUSTODIANS:

Army - GL
Air Force - 84

User activity:

Navy - EC

Preparing activity:

Army - GL

CIVIL AGENCY INTEREST:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Brushes, Dusting, Lens And Photographic Negatives H-B-1654

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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